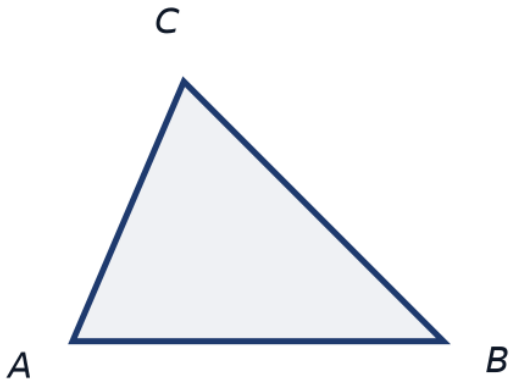
Plane Geometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

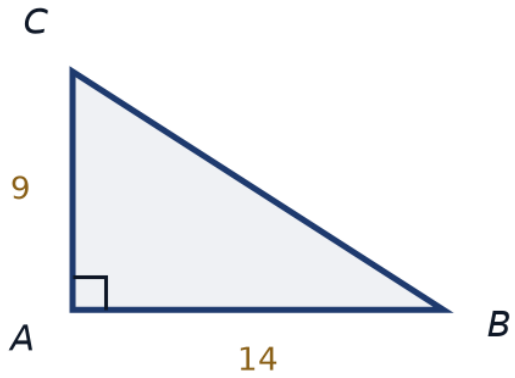
1 [Foundational] • Target: Triangle angle sum and exterior angles

A triangle has angles measuring x, $2x$, and $3x$. Find each angle.



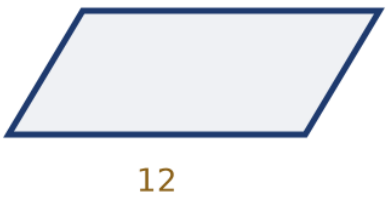
2 [Foundational] • Target: Area and perimeter of polygons

Find the area of a triangle with base 14 and height 9.



3 [Foundational] • Target: Area and perimeter of polygons

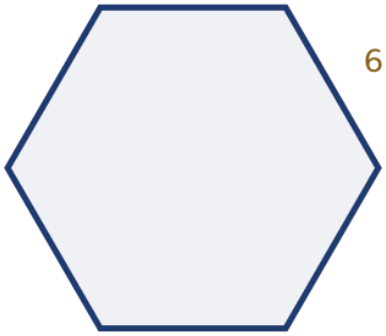
Find the area of a parallelogram with base 12 and height 5.



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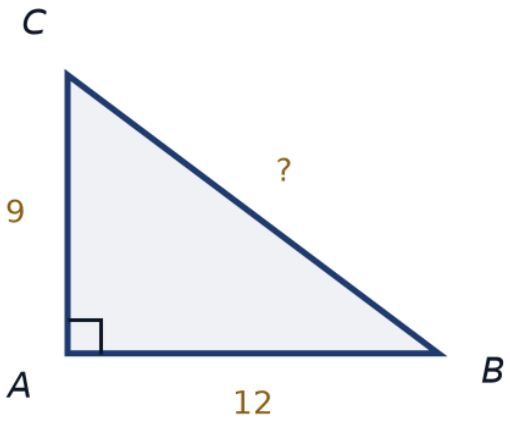
4 [Foundational] • Target: Area and perimeter of polygons

Find the area of a regular hexagon with side length 6, using the formula $A = \frac{3\sqrt{3}}{2}s^2$.



5 [Foundational] • Target: The Pythagorean theorem and special right triangles

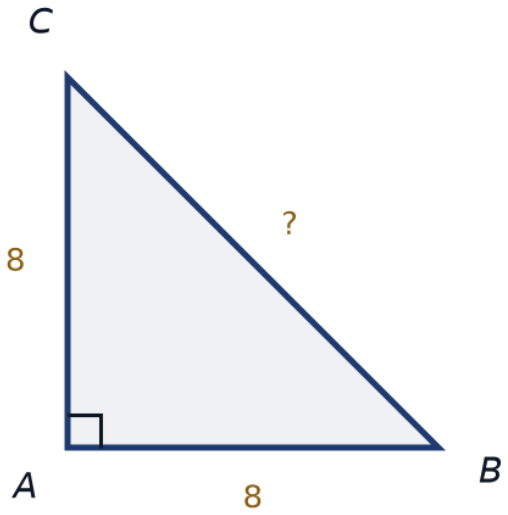
A right triangle has legs 9 and 12. Find the hypotenuse.



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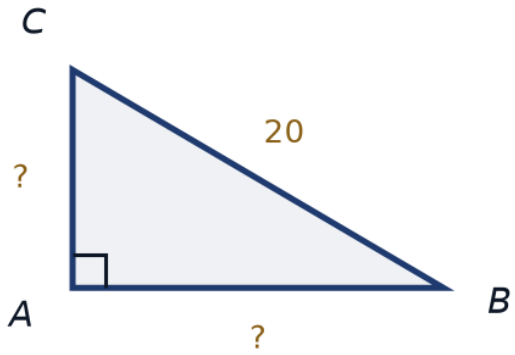
6 [Foundational] • Target: The Pythagorean theorem and special right triangles

A 45-45-90 triangle has a leg of length 8. Find the length of the hypotenuse.



7 [Foundational] • Target: The Pythagorean theorem and special right triangles

A 30-60-90 triangle has a hypotenuse of 20. Find the lengths of both legs.

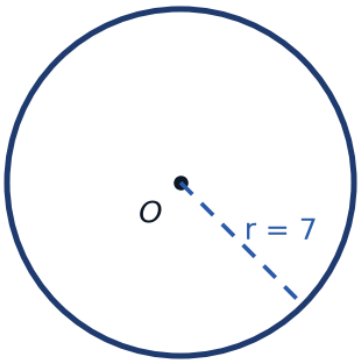


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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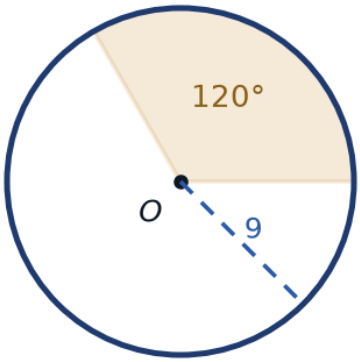
8 [Intermediate] • Target: Circles: area, circumference, arcs, and sectors

Find the area and circumference of a circle with diameter 14, in terms of π.



9 [Intermediate] • Target: Circles: area, circumference, arcs, and sectors

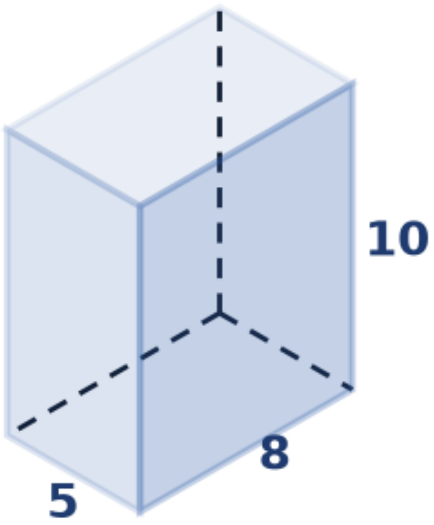
Find the arc length of a 120° sector in a circle with radius 9, in terms of π.



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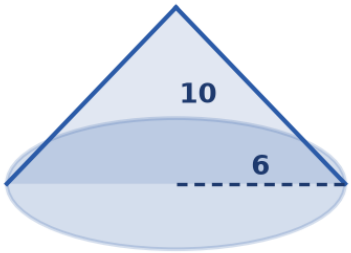
10 **[Intermediate]** • **Target: Volume and surface area of 3D solids**

Find the volume of a rectangular prism with dimensions 5, 8, and 10.



11 **[Intermediate]** • **Target: Volume and surface area of 3D solids**

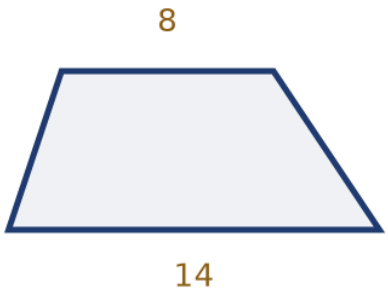
Find the volume of a cone with radius 6 and height 10, in terms of π.



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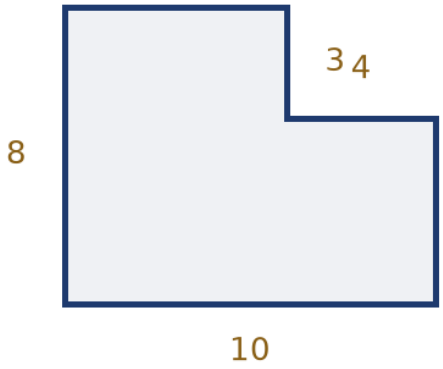
12 [Intermediate] • Target: Trapezoids and composite polygon areas

Find the area of a trapezoid with parallel sides 8 and 14, and height 6.



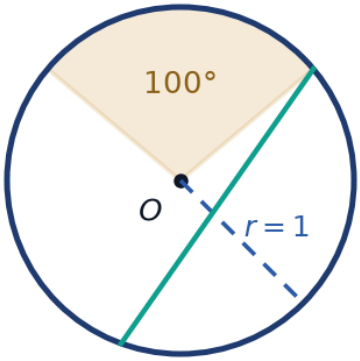
13 [Intermediate] • Target: Trapezoids and composite polygon areas

An L-shaped figure is formed by a 10-by-8 rectangle with a 4-by-3 rectangle removed from one corner. Find the remaining area.



14 [Intermediate] • Target: Chords, tangents, and inscribed angles in circles

An inscribed angle subtends an arc of 100°. Find the measure of the inscribed angle.

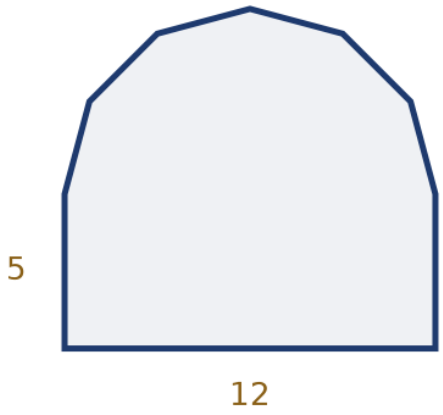


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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

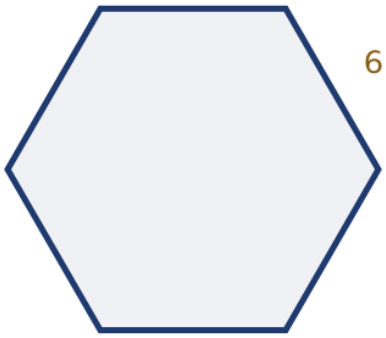
15 **[Challenging] • Target: A multi-part composite geometry problem • Cross-Lesson: 6.01 Trigonometry**

This question has two parts. A semicircular window has a diameter of 12 feet and sits atop a rectangular window of width 12 feet and height 5 feet. (a) Find the area of the semicircular portion, in terms of π. (b) Find the total area of the combined window shape.



16 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Statistics & Probability**

Find the area of a regular hexagon with side length 6, using the formula $A = \frac{3\sqrt{3}}{2}s^2$.



17 **[Challenging] • Target: Angle relationships • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers**

Two angles are supplementary, and one is 3 times the other. Find both angles.

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18 [Challenging] • Target: Angle relationships • Cross-Lesson: 1.01 Numbers, Percents & Ratios

Two parallel lines are cut by a transversal, creating an angle of 70°. Find the measure of its alternate interior angle and its corresponding angle.

19 [Challenging] • Target: Triangle angle sum and exterior angles • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

A triangle's exterior angle at one vertex measures 115°. If one of the remote interior angles is 50°, find the other remote interior angle.

20 [Challenging] • Target: Similar and congruent figures • Cross-Lesson: 3.01 Functions & Graphs

Two similar triangles have a scale factor of $\frac{3}{5}$. If the smaller triangle has area 18, find the area of the larger triangle.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Angle relationships

Two angles are complementary, and they differ by 52°. Find the larger angle.

a. 19° **b.** 66°

c. 76° **d.** 71°

22 [Foundational] • Target: Triangle angle sum and exterior angles

Two angles are complementary, and they differ by 8°. Find the larger angle.

a. 49° **b.** 44°

c. 41° **d.** 54°

23 [Foundational] • Target: Area and perimeter of polygons

Two angles are complementary, and they differ by 48°. Find the larger angle.

a. 74° **b.** 64°

c. 21° **d.** 69°

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24 [Foundational] • Target: The Pythagorean theorem and special right triangles

Two angles are supplementary, and they differ by 46°. Find the larger angle.

a. 67°

b. 108°

c. 113°

d. 118°

25 [Foundational] • Target: Circles: area, circumference, arcs, and sectors

Two angles are supplementary, and they differ by 30°. Find the larger angle.

a. 105°

b. 100°

c. 75°

d. 110°

26 [Foundational] • Target: Similar and congruent figures

Two angles are complementary, and they differ by 22°. Find the larger angle.

a. 34°

b. 61°

c. 51°

d. 56°

27 [Foundational] • Target: Volume and surface area of 3D solids

Two angles are complementary, and they differ by 76°. Find the larger angle.

a. 88°

b. 7°

c. 78°

d. 83°

28 [Foundational] • Target: Multi-step geometry problems

Two angles are complementary, and they differ by 24°. Find the larger angle.

a. 57°

b. 52°

c. 33°

d. 62°

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Trapezoids and composite polygon areas

A right triangle has legs 16 and 30. Find the length of the hypotenuse.

a. 35

b. 34

c. 33

d. 46

30 [Intermediate] • Target: Chords, tangents, and inscribed angles in circles

A right triangle has legs 6 and 8. Find the length of the hypotenuse.

a. 9

b. 11

c. 14

d. 10

31 [Intermediate] • Target: Congruence and triangle proofs

A right triangle has legs 9 and 40. Find the length of the hypotenuse.

a. 41

b. 42

c. 40

d. 49

32 [Intermediate] • Target: A multi-part composite geometry problem

A right triangle has legs 14 and 48. Find the length of the hypotenuse.

a. 50

b. 62

c. 49

d. 51

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A right triangle has legs 10 and 24. Find the length of the hypotenuse.

a. 27

b. 26

c. 25

d. 34

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A right triangle has legs 18 and 24. Find the length of the hypotenuse.

a. 42

b. 31

c. 30

d. 29

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A right triangle has legs 20 and 48. Find the length of the hypotenuse.

a. 52

b. 68

c. 53

d. 51

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A circle has radius 11. Find the arc length of a 270° sector, in terms of π.

a. $\frac{33}{2}\pi$

b. $\frac{35}{2}\pi$

c. $\frac{31}{2}\pi$

d. 33π

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A circle has radius 2. Find the sector area of a 180° sector, in terms of π.

a. 2π

b. 3π

c. 4π

d. 1π

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

A circle has radius 12. Find the arc length of a 30° sector, in terms of π.

a. 1π

b. 3π

c. 2π

d. 4π

39 [Challenging] • Target: Angle relationships • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Find the volume of a cylinder with radius 5 and height 14, in terms of π.

a. 350π

b. 360π

c. 140π

d. 340π

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40 [Challenging] • Target: Triangle angle sum and exterior angles • Cross-Lesson: 3.01 Functions & Graphs

Find the volume of a cone with radius 5 and height 10, in terms of π.

a. $\frac{244}{3}\pi$

b. $\frac{250}{3}\pi$

c. $\frac{256}{3}\pi$

d. 250π

41 [Challenging] • Target: Area and perimeter of polygons • Cross-Lesson: 4.01 Coordinate Geometry

Two similar triangles have a scale factor of $\frac{2}{5}$. If the smaller triangle has area 16, find the area of the larger triangle.

a. 105

b. 40

c. 95

d. 100

42 [Challenging] • Target: The Pythagorean theorem and special right triangles • Cross-Lesson: 6.01 Trigonometry

Two similar triangles have a scale factor of $\frac{1}{2}$. If the smaller triangle has area 1, find the area of the larger triangle.

a. 1

b. 9

c. 2

d. 4

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

Two similar triangles have a scale factor of $\frac{2}{5}$. If the smaller triangle has area 16, find the area of the larger triangle.

a. 105

b. 40

c. 95

d. 100

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

Two parallel lines are intersected by a transversal. If one interior angle measures $(3x + 15)^\circ$ and its consecutive interior angle measures $(2x + 45)^\circ$, what is the value of x?

a. 24

b. 30

c. 36

d. 40

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45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

In a 30°-60°-90° right triangle, the length of the hypotenuse is $16\sqrt{3}$. What is the length of the side opposite the 60° angle?

a. 12

b. 24

c. $8\sqrt{3}$

d. 16

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